

Secure AI-Based Handwritten Text Recognition for Indian Scripts

Handwritten documents in Indian regional languages such as Telugu, Tamil, and Hindi are still widely used for official records, academic notes, and historical archives. However, converting these handwritten materials into digital text accurately and securely remains a major challenge. Existing Optical Character Recognition (OCR) systems mainly focus on printed English or Devanagari texts and lack accuracy for diverse Indian scripts, especially for handwritten input. Moreover, current handwriting recognition solutions do not consider data privacy, model security, or tamper detection, which are crucial when dealing with personal or official handwritten data.

The proposed system, Secure AI-Based Handwritten Text Recognition for Indian Scripts, aims to develop a secure and intelligent OCR model that can accurately transcribe handwritten Indian scripts into editable digital text while ensuring privacy and integrity. The system uses deep learning models (CNN + BiLSTM + CTC) for recognizing characters and words from handwritten images. To preserve user privacy, the model will employ Federated Learning, where training occurs on local devices, and only encrypted model updates are shared with a central server—no personal data ever leaves the device. Additionally, each recognized text output will be hashed and stored on a private blockchain, ensuring that the transcribed data cannot be tampered with.

This system will particularly benefit government offices, educational institutions, digital archivists, and researchers who manage handwritten Indian records. It helps by providing faster, more accurate, and tamper-proof digitization of handwritten content while maintaining data confidentiality and trust. Unlike traditional OCR tools, this project integrates AI accuracy, cybersecurity, and privacy preservation into one unified framework tailored for Indian scripts.